

5 The Granddier's baobab is endemic to Madagascar.

Fruit bats transfer pollen from one baobab tree to another.

The population of fruit bats in Madagascar is decreasing.

The photographs show one Granddier's baobab tree in a field of rice and a Madagascan fruit bat.



(Source: © Graham, David / Alamy Stock Photo)



(Source: © John Dambik / Alamy Stock Photo)

(a) (i) This species of tree is endangered.

Identify **two** reasons why this species of tree is endangered.

Use the information above and the photographs to support your answer.

(2)

1

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2

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- (ii) Give **two** reasons how education could help conserve the Granddier's baobab.

(2)

1

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2

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- (b) Calcium ions are used to make molecules in plants.

- (i) Explain the role of **one** named molecule that uses calcium ions.

(2)

Name of molecule

Role of molecule

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.....

- (ii) Water is also important to plants.

Which of the following would transport water **down** the stem of a plant?

(1)

- ☐ **A** phloem
- ☐ **B** sclerenchyma
- ☐ **C** vacuole
- ☐ **D** xylem

- (c) The photograph shows pods on a soybean plant.



pod

(Source: © B Christopher / Alamy Stock Photo)



The effect of magnesium ions on the number of pods per soybean plant was investigated.

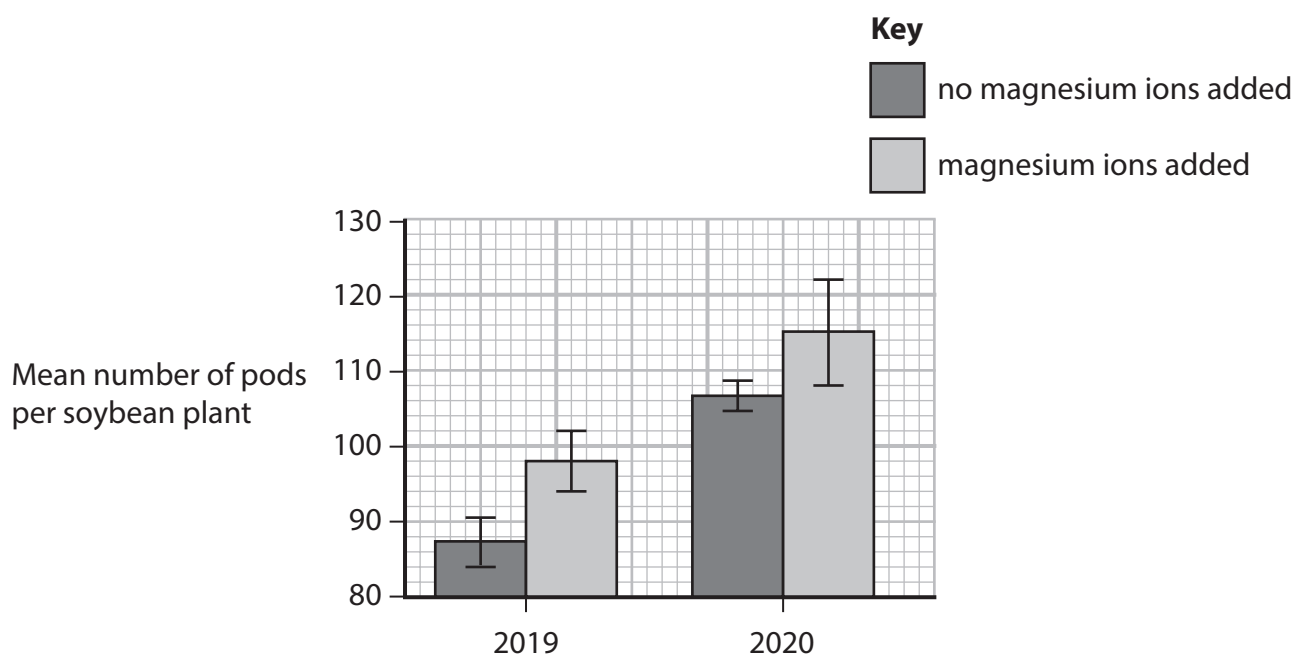
In 2019, soybean plants were grown in two fields.

One field had no magnesium ions added to the soil and the other field had magnesium ions added to the soil.

The mean number of pods per soybean plant was determined for both fields.

In 2020, the investigation was repeated using the same fields.

The graph shows the results of this investigation.



Give **two** conclusions that can be made using this data.

(2)

1

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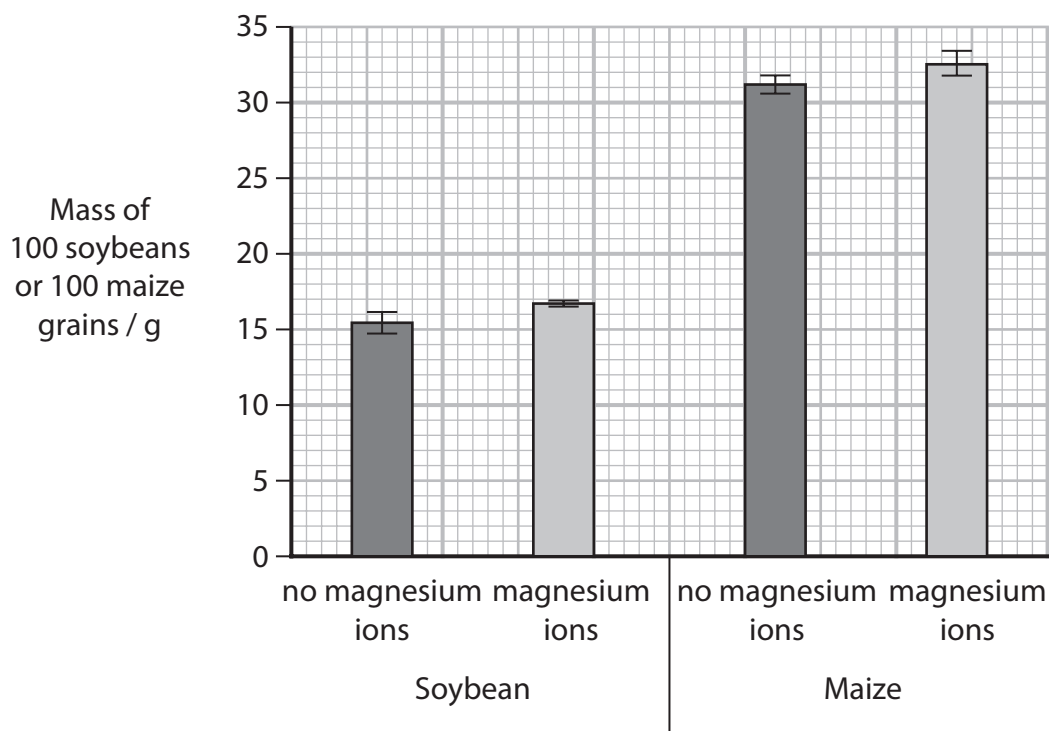
2

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- (d) In another investigation, the effect of magnesium ions on the mass of soybeans and maize grains was studied.

The graph shows the results of this investigation.



Comment on the results from this investigation.

(4)

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(Total for Question 5 = 13 marks)

